

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS

9188/4

PAPER 4 Practical Test

NOVEMBER 2015 SESSION

2 hours 30 minutes

Candidates answer on the question paper.

Additional materials:

As listed in Instructions to Supervisors

Electronic calculator

Graph paper

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page and on any separate answer paper used.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

In Questions 1 and 2, you are expected to record all your observations as soon as these observations are made, and to plan the presentation of the records so that it is not necessary to make a fair copy of them. The working of the answers is to be handed in. Marks are mainly given for a clear record of the observations actually made, for their suitability and accuracy, and for the use made of them. **Routine** precautions and theory are **not** wanted in Questions 1 and 2. You should, however, record any **special** precautions you have taken so as to aid accuracy. At the end of the examination, fasten any separate answer paper used securely to the question paper.

INFORMATION FOR CANDIDATES

Questions 1 and 2 carry 18 marks each and question 3 carries 14 marks.

Squared paper and Mathematical tables are available.

Additional paper and graphs should be submitted **only** if it becomes **necessary** to do so.

You are advised to spend approximately one hour on each of Questions 1 and 2 and 30 minutes on Question 3

FOR EXAMINER'S USE	
1	
2	
3	
TOTAL	

This question paper consists of 9 printed pages, 2 lined pages and 1 blank page.

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- 1 In this experiment you will investigate the phase relationship between two torsional oscillating systems.

The torsional system A in Fig.1.1 has been set up for you and should **not** be changed throughout the experiment.

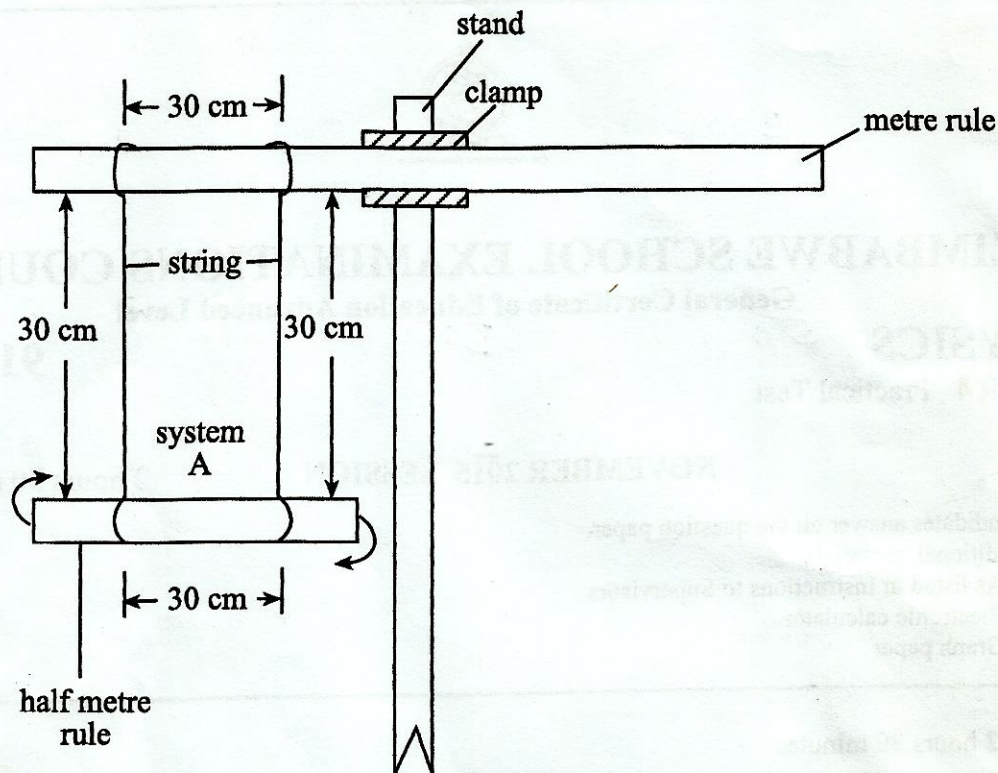


Fig.1.1

- (a) (i) Set up torsional system B, on the other side of the metre rule as shown in Fig. 1.2.

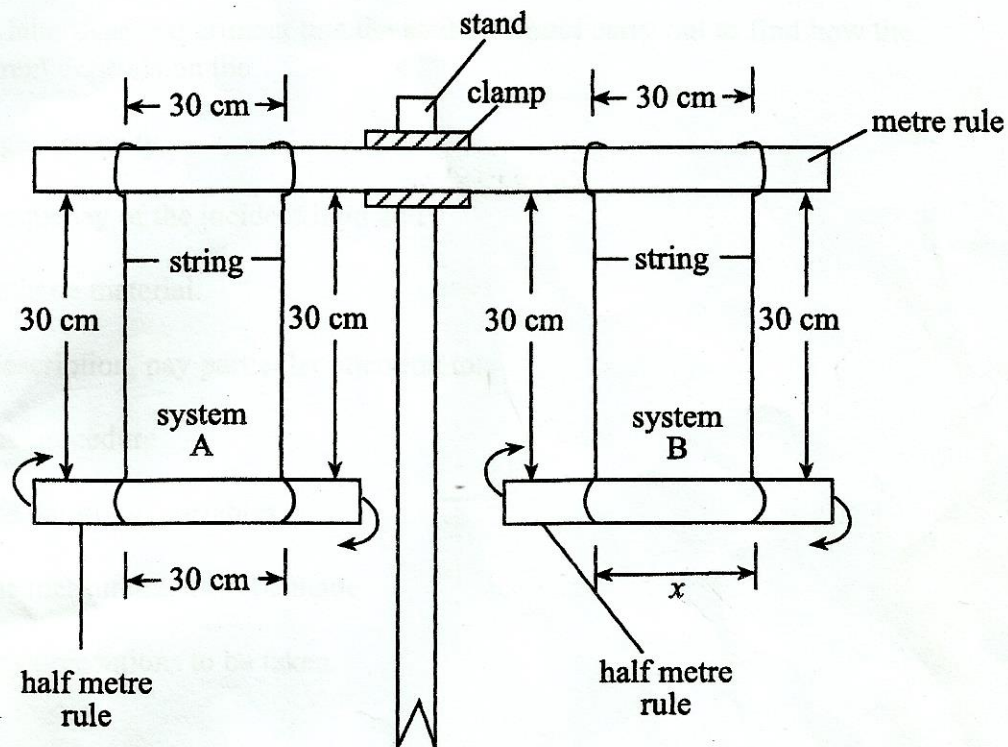


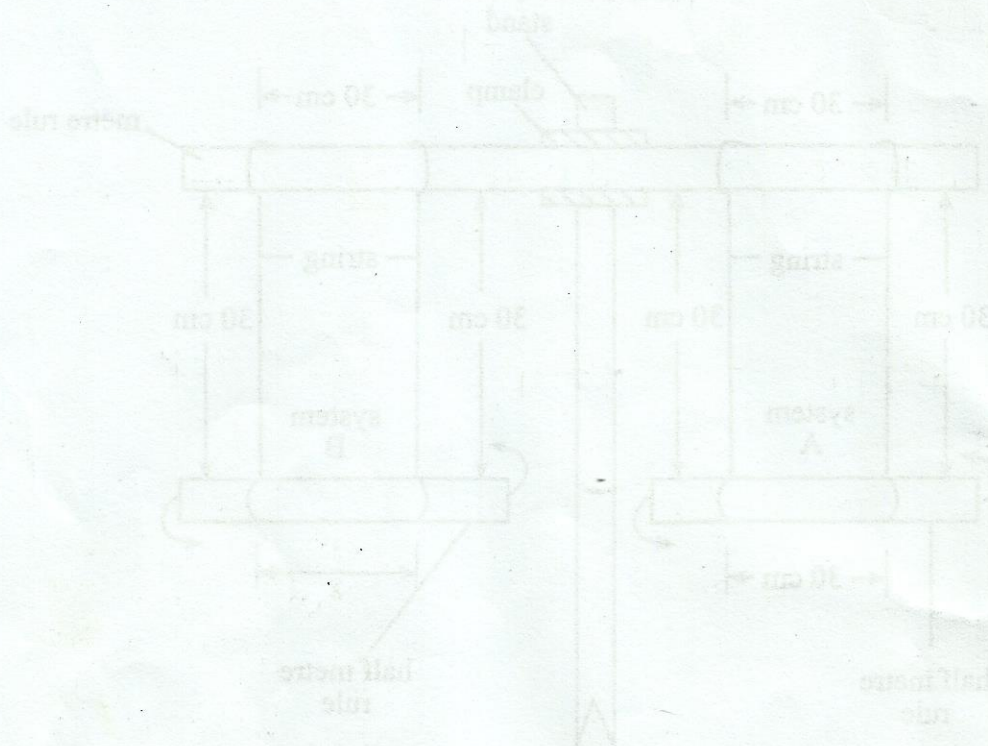
Fig.1.2

- (ii) With distance x of 45 cm, displace both systems so that they perform torsional oscillations in a horizontal plane. You will observe that the systems will at one point be in phase (move in same direction), go out of phase and come back in phase.
- (iii) Measure and record the time, t , from the moment the systems are in phase to the moment they are both in phase again.
- (iv) Change the value of x and repeat a(iii) for seven more sets of readings for x in the range $45 \text{ cm} \geq x \geq 15 \text{ cm}$. Include values of $\log t$ and $\log x$ in your table of results.

Calculate the percentage error in the value of t at $x = 40 \text{ cm}$.

- (v) Justify the number of significant figures used for $\log x$.
- (b) (i) Plot a graph $\log t$ (y -axis) against $\log x$ (x -axis).
- (ii) Use the graph to determine the greatest time taken for the system to be in phase and return in phase. Include the corresponding value of x .

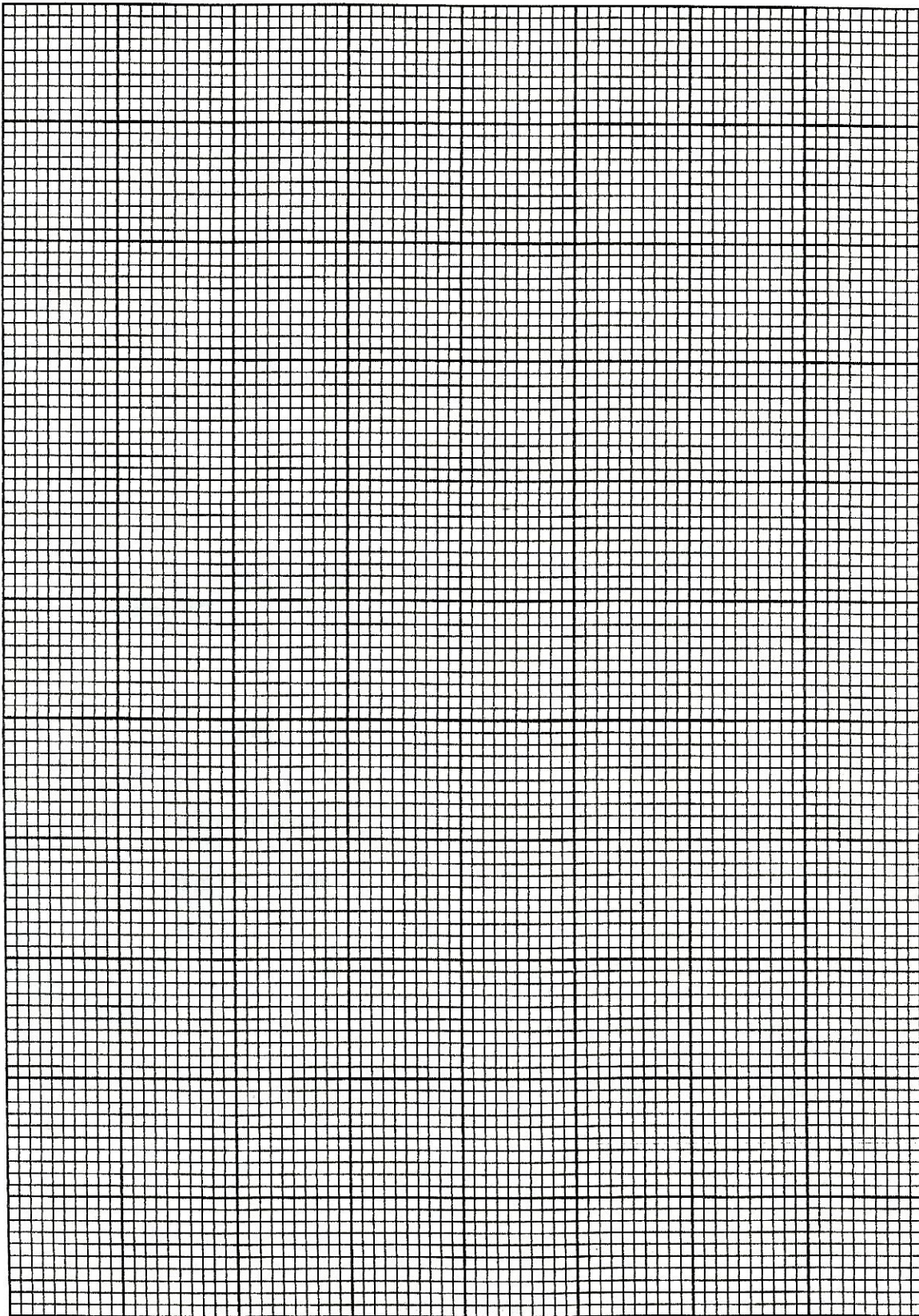
Measurements and calculations



M

R

A



G

- 2 In this experiment you will investigate the variation of the voltage across a wire with the current-length of the wire.

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- (a) (i) Measure the diameter of wire X.
(ii) Calculate the cross sectional area of wire X.

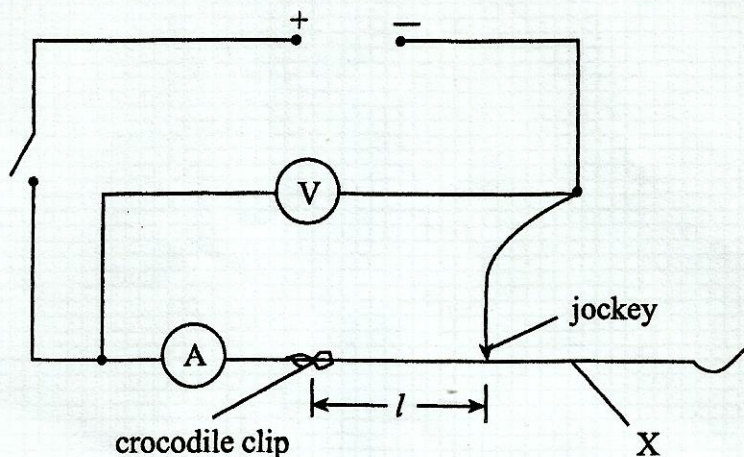


Fig.2.1

- (b) (i) Set up the circuit shown in Fig.2.1.
(ii) With length $l = 100$ mm, close the switch and record the current, I , voltage, V , and length, l .
(iii) Increase l by about 100 mm. Close the switch and record I , V and l .
(iv) Repeat (iii) to obtain four further values of I and V for l in the range $200 \text{ mm} < l \leq 600 \text{ mm}$.
(v) Include values of $I \times l$ in your table of results.

Justify the number of significant figures used for $I \times l$.

- (c) It is suggested that V and $I \times l$ are related by the equation

$$V = \frac{5.1 \times 10^{-7}}{A} (I \times l) + B$$

where A and B are constants.

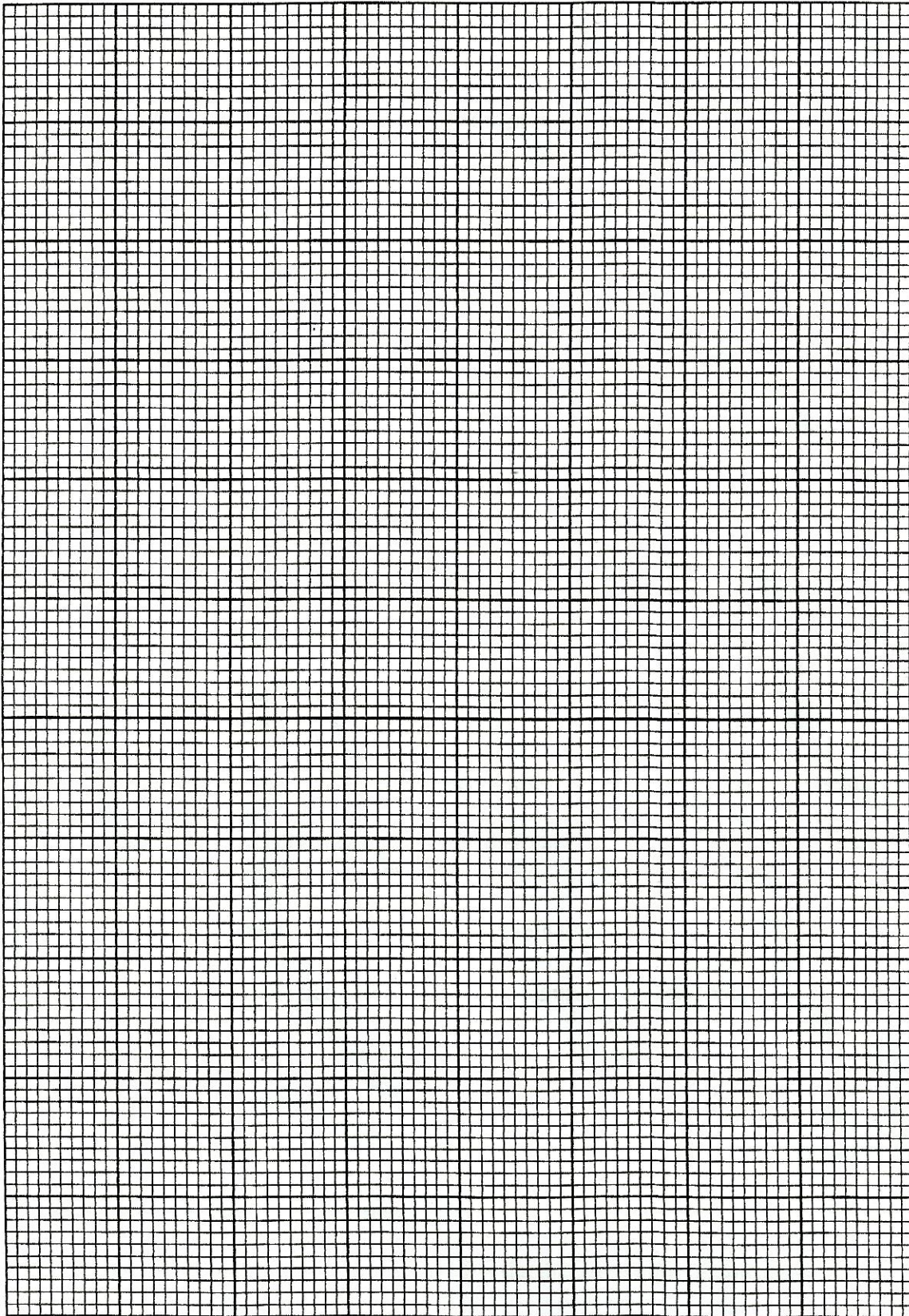
- (i) Plot a suitable graph to enable the determination of constants A and B .
(ii) From the graph determine A and B .
(d) Compare the value of A to the cross-sectional area calculated in (a).

Measurements and calculations

M

R

A



G

3

A student wishes to investigate factors which affect the magnitude of the photocurrent in a photoelectric experiment.

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Design a laboratory experiment that the student would carry out to find how the photocurrent depends on the

- (i) light intensity,
- (ii) frequency of the incident light and
- (iii) cathode material.

In your description, pay particular attention to:

- (a) the procedure
- (b) the control of variables
- (c) the measurements to be made
- (d) any precautions to be taken

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